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Mermaid Live Editor <https://mermaid.live>

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```
erDiagram
    users ||--o{ referral_codes : "████(referrer_id)"
    users ||--o{ referral_usage_logs : "██████"
    users ||--o{ referral_relationships : "██████(referrer_id)"
    users ||--o{ referral_relationships : "██████(referee_id)"
    users ||--o{ commission_records : "████(user_id)"

    referral_codes ||--o{ referral_usage_logs : "████(code)"
    referral_codes {
        int id PK
        string code UK "██████"
        string code_type "██:creator/public_welfare/..."
        int referrer_id FK "████ID"
        string referred_by_code "██████"
        string status "active/inactive/depleted"
        int use_count "████"
        int max_uses "██████"
        string expires_at "████"
        string batch_id "██ID"
        text remark "██"
    }

    referral_usage_logs {
        int id PK
```

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        string code FK "####"
        int user_id FK "####ID"
        string scene "##:register/manual_bind"
        string created_at
    }

    referral_relationships {
        int id PK
        int referrer_id FK "####ID"
        string referrer_code "#####"
        int referee_id FK "####ID"
        string referee_code "#####"
        string status "active/inactive"
        int level "##:1=##,2=##"
        decimal total_commission "####"
        text remark
    }

    commission_records {
        int id PK
        int user_id FK "####ID"
        int referral_relationship_id FK "#####"
        string type "##:register/upgrade/purchase"
        decimal amount "####"
        string status "pending/confirmed/paid"
        string paid_at "#####"
    }

    user_referrals {
        int id PK
        int referrer_id FK
        int referee_id FK
        string referral_code
        string status
    }

```

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```

sequenceDiagram
    participant Admin as ■■■
    participant Frontend as ■■■■
    participant Backend as ■■API
    participant DB as ■■■

    Admin->>Frontend: ■■"#####"■■
    Frontend->>Backend: GET /api/admin/referral-codes/list
    Backend->>DB: SELECT * FROM referral_codes
    DB->>Backend: #####
    Backend->>Frontend: {code:0, data: [...]}
    Frontend->>Frontend: #####Picker■■■

    Admin->>Frontend: ##### + #####
    Admin->>Frontend: ■■"■■"

    Frontend->>Frontend: ■■■■
    Frontend->>Backend: POST /api/admin/referral-codes/bind
    Note over Frontend,Backend: {referrer_code: "LCRG48KL", referred_code: "GYRG..."

    Backend->>Backend: #####
    Backend->>DB: SELECT * FROM referral_codes WHERE code = ?
    DB->>Backend: #####

    alt #####
        Backend-->>Frontend: {code:-1, message:"#####"}
        Frontend-->>Admin: ■■■■
    end

    Backend->>DB: SELECT * FROM referral_codes WHERE code = ?
    DB->>Backend: #####

```

```

alt ██████████
  Backend-->>Frontend: {code:-1, message:"██████████"}
  Frontend-->>Admin: ██████
end

Backend-->>DB: CHECK referred_by_code IS NOT NULL

alt ██████
  Backend-->>Frontend: {code:-1, message:"██████████"}
  Frontend-->>Admin: ██████
end

Backend-->>DB: BEGIN TRANSACTION
DB-->>Backend: ██████

Backend-->>DB: UPDATE referral_codes SET referred_by_code = ? WHERE code = ?
DB-->>Backend: ██████

Backend-->>DB: INSERT INTO referral_relationships (...)
DB-->>Backend: ██████

Backend-->>DB: COMMIT
DB-->>Backend: ██████

Backend-->>Frontend: {code:0, message:"██████████"}
Frontend-->>Admin: ███"██████"
Frontend-->>Frontend: ██████████

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graph TD
  Start([Start([████████])]) --> TriggerType{TriggerType{██████?}}

  TriggerType -->|██████| Calc1[██████████]
  TriggerType -->|██████████| Calc2[██████████]
  TriggerType -->|██████████| Calc3[██████████]
  TriggerType -->|██| Calc4[██████████]

  Calc1 --> GetReferral[██████████]
  Calc2 --> GetReferral
  Calc3 --> GetReferral
  Calc4 --> GetReferral

  GetReferral --> CheckLevel{CheckLevel{██████?}}

  CheckLevel -->|██████| Rate1[██20%]
  CheckLevel -->|██████| Rate2[██15%]
  CheckLevel -->|██████| Rate3[██10%]
  CheckLevel -->|██████| Rate4[██12%]
  CheckLevel -->|██████| Rate5[██25%]

  Rate1 --> CheckSub[██████████]
  Rate2 --> CheckSub
  Rate3 --> CheckSub
  Rate4 --> CheckSub
  Rate5 --> CheckSub

  CheckSub -->|███| CalcSubCommission[██████████]
  CheckSub -->|███| GenerateRecord[██████████]

  CalcSubCommission --> SubRate{SubRate{██████}}
  SubRate -->|██| SubRate1[██5%]
  SubRate -->|██| SubRate2[██3%]
  SubRate -->|██| SubRate3[██2%]

  SubRate1 --> GenerateRecord
  SubRate2 --> GenerateRecord
  SubRate3 --> GenerateRecord
  GenerateRecord --> InsertCommission[INSERT INTO commission_records]

```



```

        rr.level,
        t.depth + 1
    FROM referral_relationships rr
    INNER JOIN team t ON rr.referrer_id = t.referee_id
    WHERE t.depth < 5 -- ■■■5■
)
SELECT * FROM team;

```

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sql
SELECT
    u.id,
    u.nickname,
    rc.code as referral_code,
    COUNT(DISTINCT rr.referee_id) as total_referees,
    COUNT(DISTINCT CASE WHEN u_reg.created_at >= date('now', '-30 days') THEN rr...
    COALESCE(SUM(cr.amount), 0) as total_commission
FROM users u
LEFT JOIN referral_codes rc ON rc.referrer_id = u.id
LEFT JOIN referral_relationships rr ON rr.referrer_id = u.id
LEFT JOIN commission_records cr ON cr.user_id = u.id AND cr.status = 'paid'
WHERE u.role IN ('creator', 'professional', 'public_welfare')
GROUP BY u.id;

```